

GYM MANAGEMENT SYSTEM

(Mini Project)

✧ Introduction :

The Gym Management System is a simple database project developed using MySQL. It is designed to store and manage information related to gym members, trainers, membership plans, memberships, and payments. The project demonstrates important Database Management System concepts such as tables, primary keys, foreign keys, constraints, relationships, normalization, CRUD operations, SQL joins, views, triggers, and transactions. The system is developed only at the database level and does not include any graphical or web interface.

✧ Problem Statement :

Managing gym records manually can make it difficult to maintain information about members, trainers, memberships, and payments. Manual records can also result in duplicate data, difficulty in searching records, and errors while maintaining payment and membership information. The proposed Gym Management System provides a structured relational database using MySQL to store and manage gym-related information efficiently.

✧ Objectives :

The main objectives of the Gym Management System are:

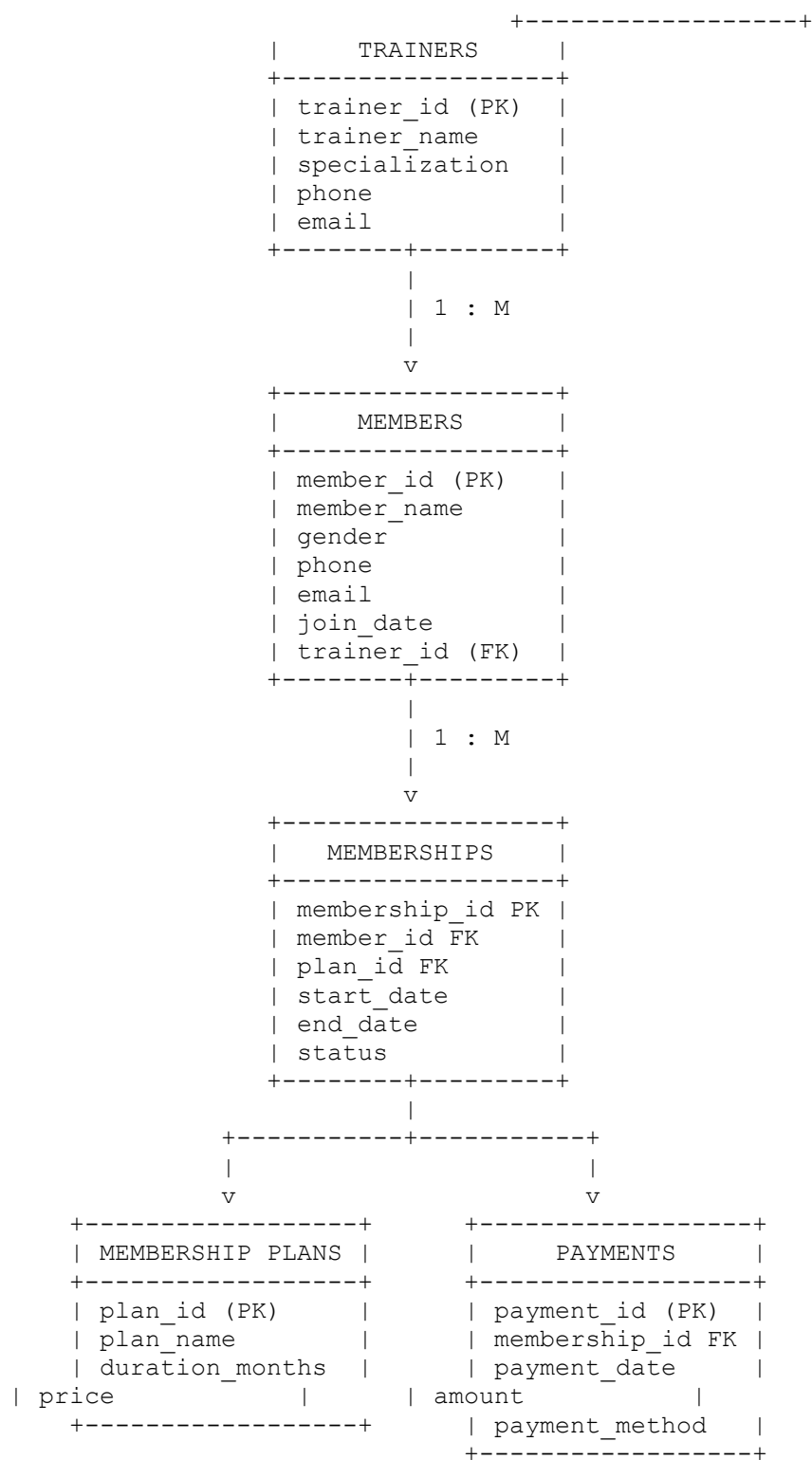
1. To design a simple relational database for gym management.
2. To store information about gym members and trainers.
3. To manage different membership plans.
4. To maintain membership and payment records.
5. To demonstrate primary keys and foreign keys.
6. To perform CRUD operations using SQL.
7. To use SQL joins and aggregate functions.
8. To demonstrate a view, trigger, and transaction.
9. To maintain data consistency using database constraints.

✧ Technologies used :

Technology	Purpose
MySQL	Database Management System
SQL	Creating and manipulating the database
MySQL Workbench	Creating and executing SQL queries

✧ ER Diagram :

The basic relationship between the tables is:



❖ Database Schema :

1. Table Name: trainers

Attribute	Data Type	Key/Constraint
trainer_id	INT	Primary Key
trainer_name	VARCHAR(100)	NOT NULL
specialization	VARCHAR(100)	—
phone	VARCHAR(15)	UNIQUE
email	VARCHAR(100)	UNIQUE

2. Table Name: members

Attribute	Data Type	Key/Constraint
member_id	INT	Primary Key
member_name	VARCHAR(100)	NOT NULL
gender	ENUM	NOT NULL
phone	VARCHAR(15)	UNIQUE
email	VARCHAR(100)	UNIQUE
join_date	DATE	NOT NULL
trainer_id	INT	Foreign Key

3. Table Name: membership_plans

Attribute	Data Type	Key/Constraint
plan_id	INT	Primary Key
plan_name	VARCHAR(50)	UNIQUE, NOT NULL
duration_months	INT	CHECK
price	DECIMAL(10,2)	CHECK

4. Table Name: memberships

Attribute	Data Type	Key/Constraint
membership_id	INT	Primary Key
member_id	INT	Foreign Key
plan_id	INT	Foreign Key
start_date	DATE	NOT NULL
end_date	DATE	NOT NULL
status	ENUM	DEFAULT

✧ SQL Queries and Features Implemented :

The following DBMS concepts are implemented in the project:

DDL

- CREATE DATABASE
- CREATE TABLE

DML

- INSERT
- UPDATE
- DELETE

DQL

- SELECT

Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- CHECK
- DEFAULT

Other SQL Features

- INNER JOIN
- LEFT JOIN
- GROUP BY
- COUNT()
- SUM()
- VIEW
- TRIGGER
- TRANSACTION
- COMMIT
- ROLLBACK

✧ Screenshot :

MySQL Workbench

SQL Editor | gym_management

SELECT m.member_id, m.member_name, m.phone, t.trainer_name FROM members m LEFT JOIN trainers t ON m.trainer_id=t.trainer_id; Query OK • 0.00 sec

member_id	member_name	phone	trainer_name
1	Aarav Joshi	9000000001	Rahul Sharma
2	Sneha Kulkarni	9000000002	Priya Patil
3	Rohan Patil	9000000003	Amit Verma
4	Ananya Deshmukh	9000000004	Priya Patil
5	Vivek Singh	9000000005	Rahul Sharma

5 row(s) returned

MySQL Workbench

SQL Editor | gym_management

SELECT m.member_name, p.plan_name, p.duration_months, ms.start_date, ms.end_date, ms.status ...; Query OK • 0.00 sec

member_n	plan_name	duration_months	start_date	end_date	status
Aarav Joshi	Standard	3	2026-01-10	2026-04-09	Expired
Sneha Kulk	Premium	6	2026-02-05	2026-08-04	Expired
Rohan Patil	Basic	1	2026-03-12	2026-04-11	Expired
Ananya Des	Standard	3	2026-04-15	2026-07-14	Expired
Vivek Singh	Premium	6	2026-05-20	2026-11-19	Active

5 row(s) returned

MySQL Workbench

SQL Editor | gym_management

SELECT py.payment_id, m.member_name, py.payment_date, py.amount, py.payment_method ...; Query OK • 0.00 sec

payment_id	member_name	payment_date	amount	payment_method
1	Aarav Joshi	2026-01-10	2700.00	UPI
2	Sneha Kulkarni	2026-02-05	4800.00	Card
3	Rohan Patil	2026-03-12	1000.00	Cash
4	Ananya Deshmukh	2026-04-15	2700.00	UPI
5	Vivek Singh	2026-05-20	4800.00	Card

5 row(s) returned

MySQL Workbench

SQL Editor | gym_management

SELECT payment_method, SUM(amount) AS total_amount FROM payments GROUP BY payment_method; Query OK • 0.00 sec

payment_method	total_amount
Card	9600.00
Cash	1000.00
UPI	5400.00

3 row(s) returned

✧ Conclusion :

The Gym Management System successfully demonstrates the implementation of a relational database using MySQL. The project includes important DBMS concepts such as tables, primary keys, foreign keys, constraints, relationships, normalization, CRUD operations, joins, aggregate functions, views, triggers, and transactions.

The system provides a simple and organized method for storing and managing gym members, trainers, membership plans, memberships, and payment information. Although this project contains only the database layer, it can be extended into a complete gym management application in the future.